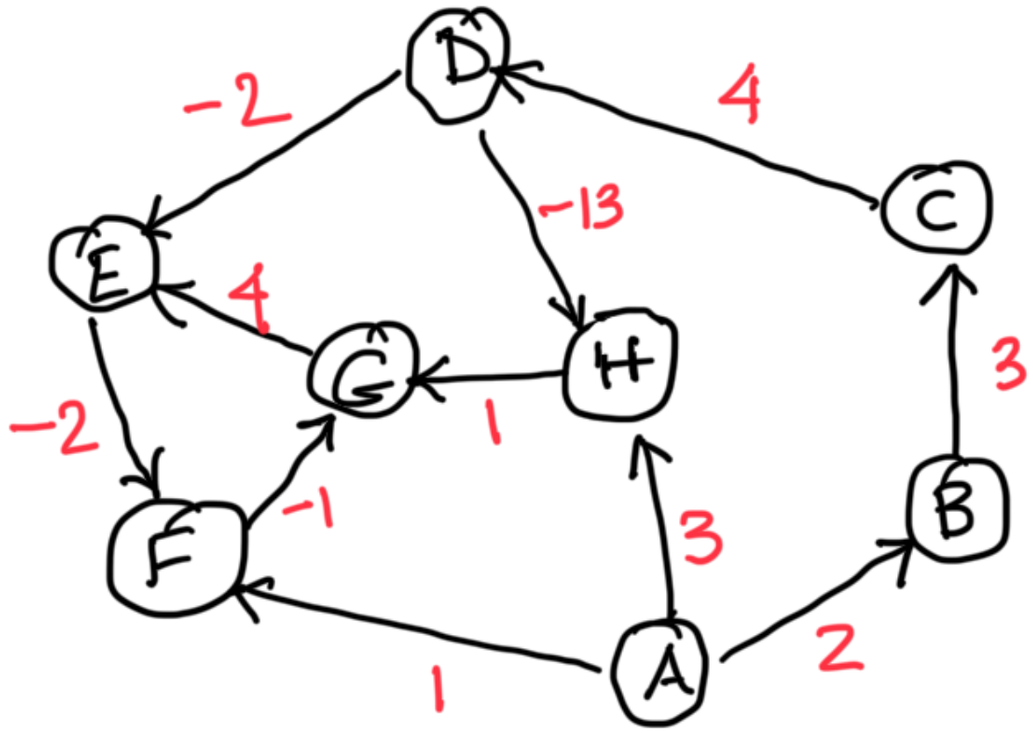
Lecture 21, DAA April 3

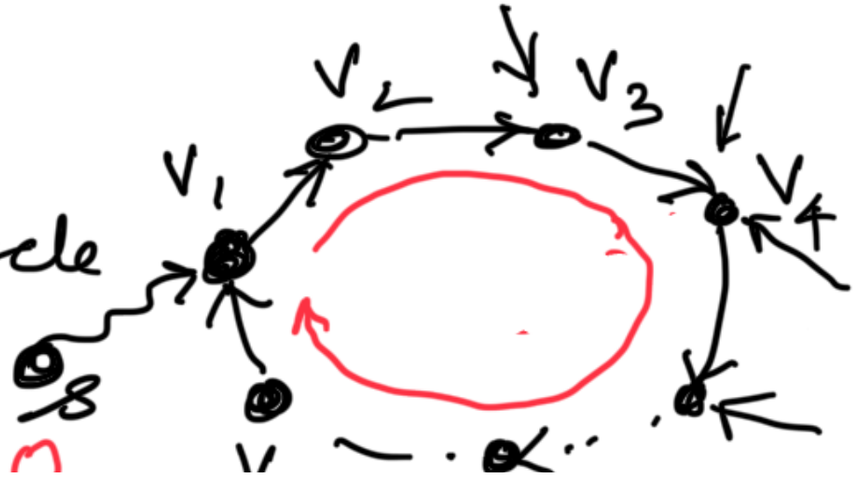


Terminating condition for Bellman Ford

Claim If $\Delta(v)$ decreases in the n^{th} iteration

$\Leftrightarrow$ negative cycle

Proof Consider any negative cycle



$w(v_1, v_2) + w(v_2, v_3) + \cdots w(v_k, v_1) = -\alpha < 0$

v_k $k \leq n-1$

Relaxation sequence over consecutive iterations

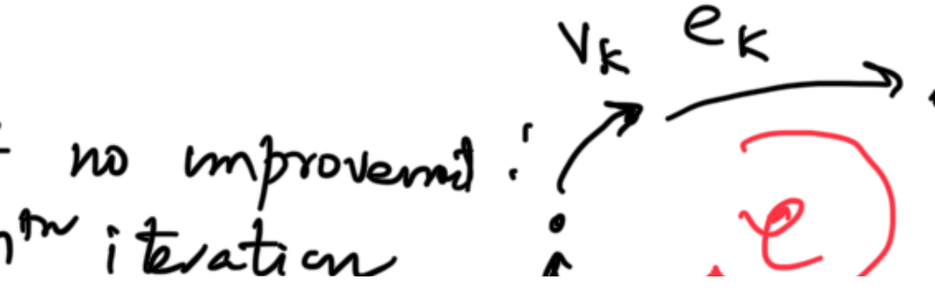
Relax(v_1, v_2), Relax(v_2, v_3) ---- Relax(v_k, v_1) will result in $\Delta(v_1)$ being lowered by at least $-\alpha$ after k iterations

$\Rightarrow$ $\Delta(v_2)$ getting lowered by $-\alpha$ after $k+1$

$\Rightarrow$ $\Delta(v_3)$ " " etc. $\Rightarrow$ Every successive iteration, some $\Delta(v_i)$ will be lowered including n^{th} iter.

Terminating condition for BF is run for n iterations. If no improvement for any $\Delta(v)$ then no $-ve$ cycle

If no improvement in n^{th} iteration



Consider any cycle C

$\Delta^n(v_i) = \Delta^{n-1}(v_i)$

$\Delta^n(v_i) \leq \Delta^{n-1}(v_{i-1}) + \omega(v_{i-1}, v_i)$

no relaxation)

$= v_1, v_2 \ldots v_k$

$$\Delta^n(v_k) \leq \Delta^{n-1}(v_{k-1}) + \omega(v_{k-1}, v_k)$$

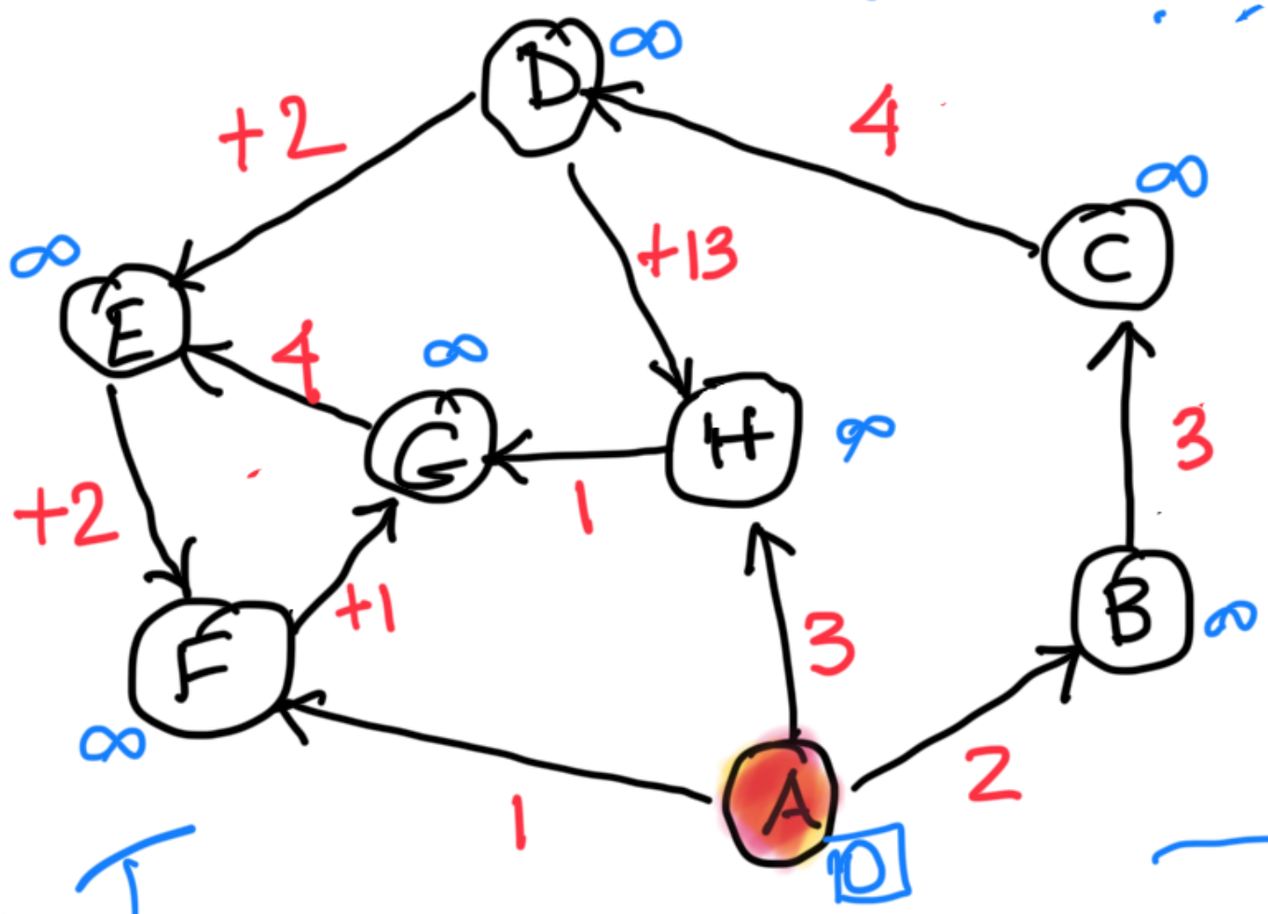
$$\Delta^{\cdot}(v_{k-1}) \leq \Delta^{\cdot}(v_{k-2}) + \omega(v_{k-2}, v_{k-1})$$

$$\vdots$$

$$\Delta^n(v_1) \leq \Delta^{n-1}(v_k) + \omega(v_k, v_1)$$

Adding $\sum \Delta^n(v_i) \leq \sum \Delta^{n-1}(v_i) + \omega(C)$

NO NEGATIVE WEIGHTS!?



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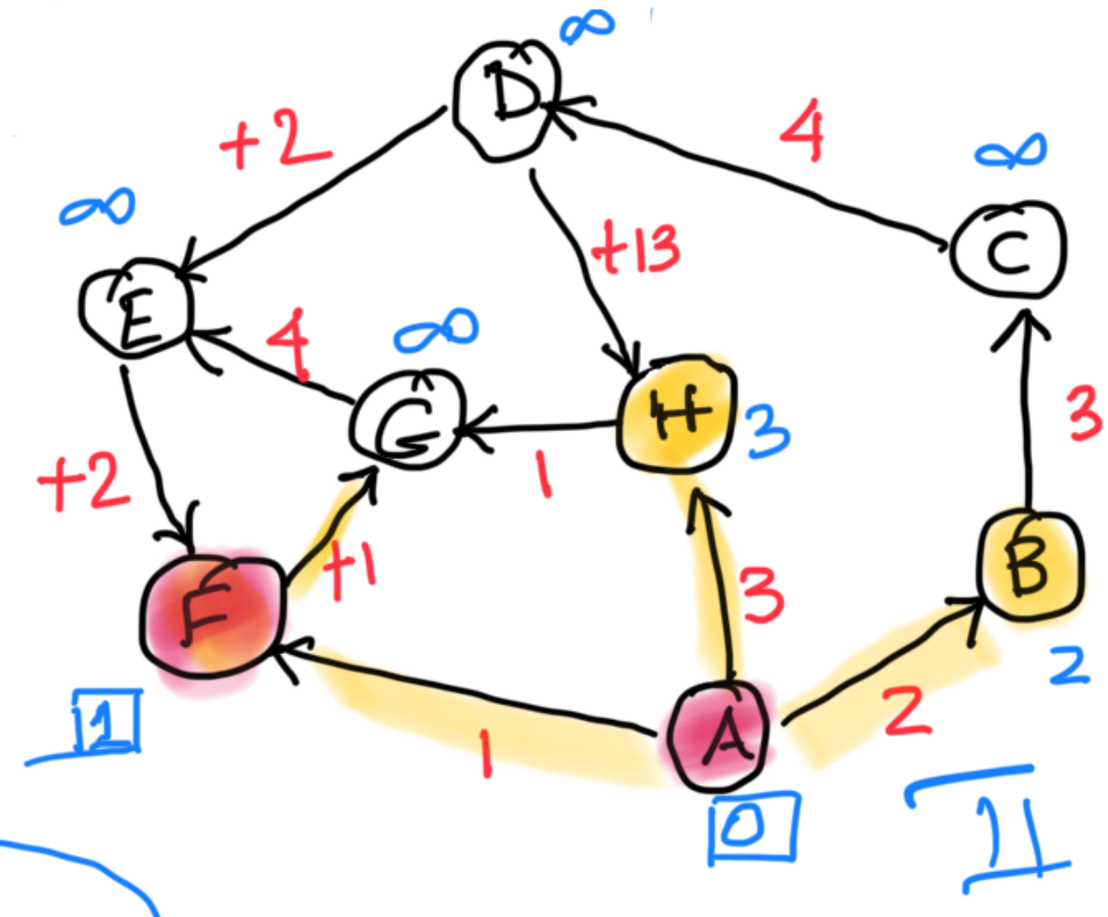

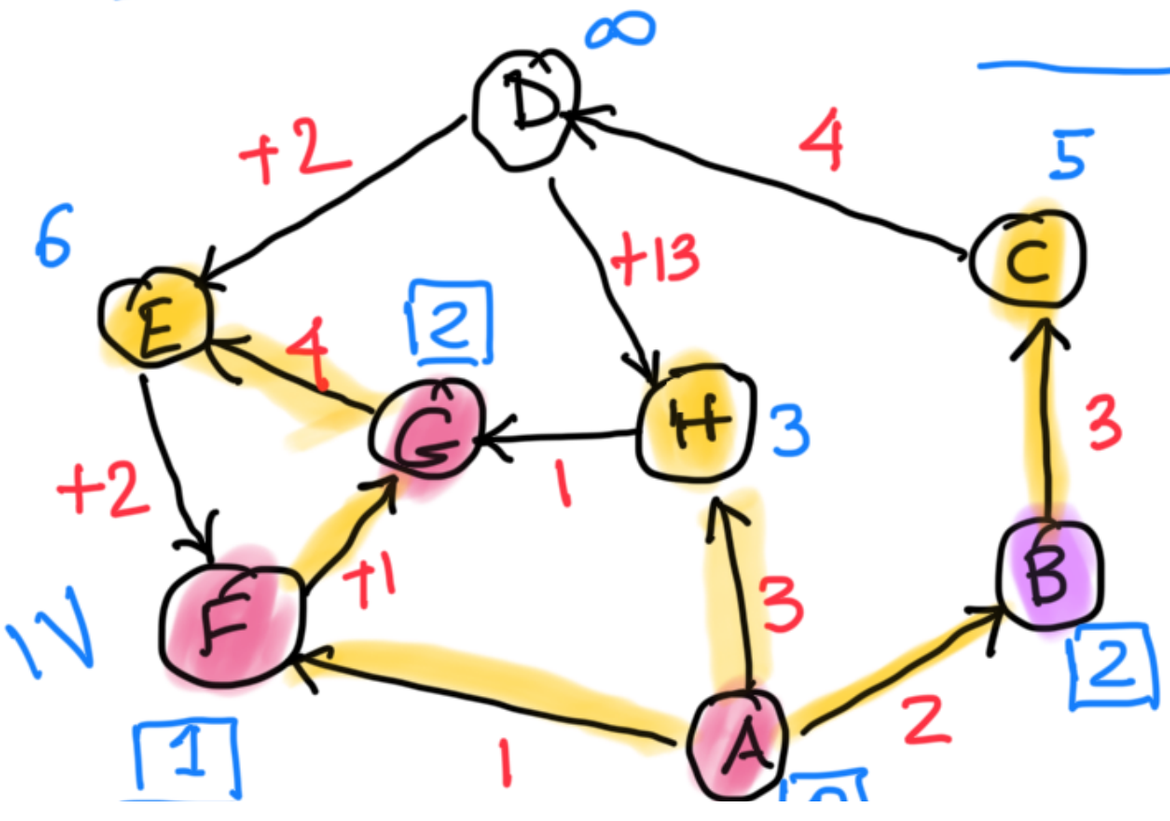

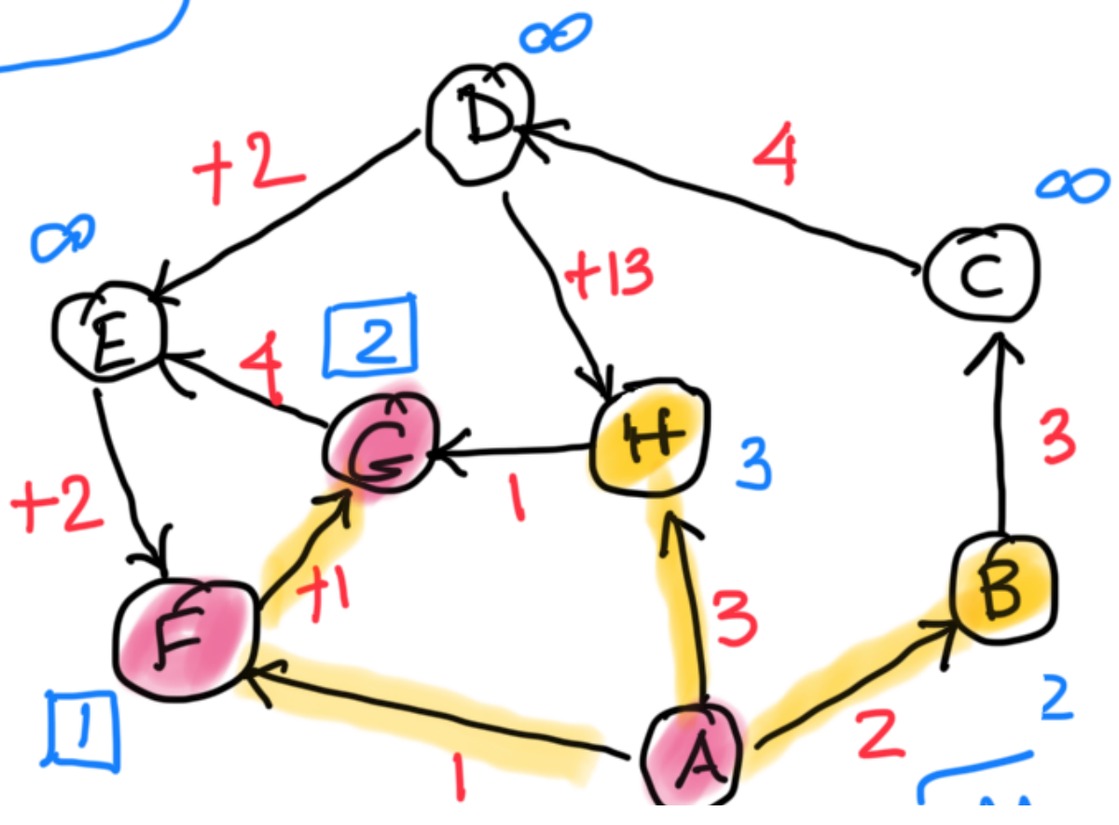

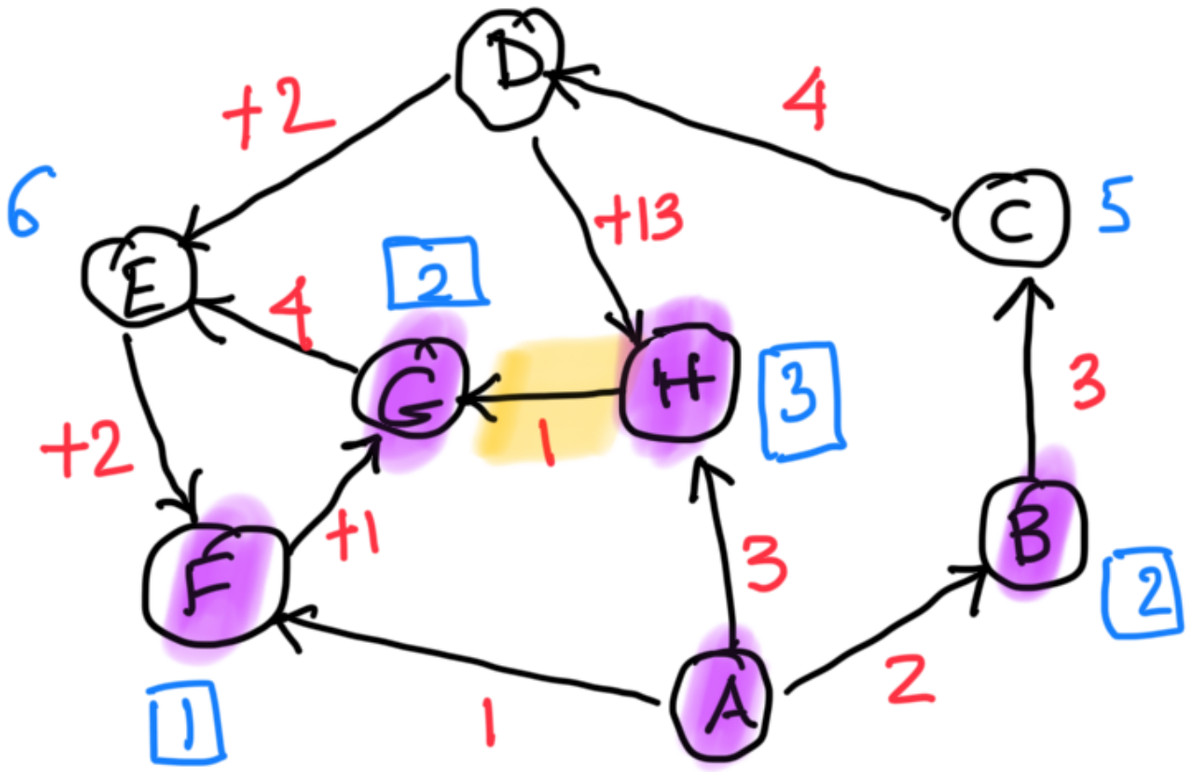

Dijkstra's Algorithm SSSP (non-negative wts)

V: Set of vertices

U: set of active vertices

S : set of frozen vertices

Initialize $S = \{s\}$ $\mathcal{U} = V - \{s\}$

$\Delta(s) = \delta(s) = 0$ $\Delta(v) = \infty$ $v \in V - \{s\}$

latest = s

While $\mathcal{U} \neq \phi$ repeat

- Relax (latest, w) $\forall w \in$ neighbors of latest
- latest := $\underset{v \in \mathcal{U}}{\operatorname{argmin}} \Delta(v)$
- δ (latest) = Δ (latest)
- $S := S \cup \{\text{latest}\}$ latest is frozen

Output $\delta(v)$ of $v \in S$

Running Time : $O((|E|+|V|)\log|V|)$

using min heaps

Correctness :

Claim : $\forall w$, when w is moved into S from U

$$\delta(w) = \Delta(w)$$

Proof by contradiction : Suppose in a run of the algorithm, w is the first vertex for which $\Delta(w) \neq \delta(w)$ and $\Delta(w)$ is minimum

All the previously frozen vertices have $\delta(v_i) = \Delta(v_i)$

is a shortest path

① $\boxed{\Delta(v_k) > \delta(v_k)}$ when it was frozen

when v_i was frozen (v_i, v_{i+1}) would be relaxed in the subsequent iteration

← this was correct

$$\Delta(v_{i+1}) = \delta(v_i) + w(v_i, v_{i+1})$$

$$= \delta(v_{i+1})$$ (since it is on the shortest path to w

$$\leq \delta(v_k)$$ non negative wts

So $\Delta(v_k) > \delta(v_k) \geq \Delta(v_{i+1})$ and v_k shouldn't have been chosen

Contradiction that $\Delta(w)$ was minimum

Useful property of Dijkstra's algorithm

The vertices are added to S (frozen) in order of increasing distances from the starting vertex